

DSI 25 – Nov 23, 2025

DASI Data Analytics and Statistics Instruction (DASI) Topics and Tools for Improved and Engaged Learning

Using LucidChart/Spark for Student Collaboration

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bit.ly/KKPresentationFiles

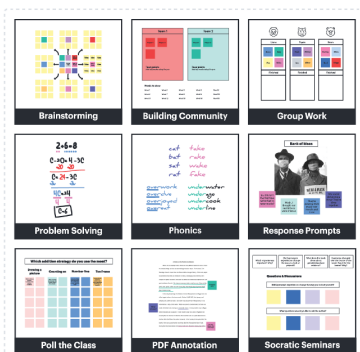
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Lucid for Education is a suite of instructional tools

Every Canvas instructor and student gets a license to the *entire Lucid for Education Suite* for free

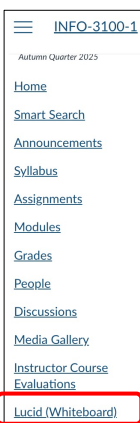
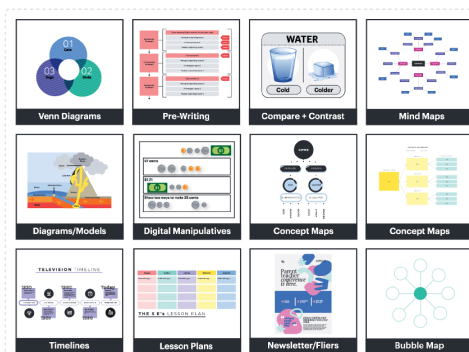
*Lucidspark

Collaborative virtual whiteboard



Lucidchart

Interactive diagramming and graphic organizers



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Lucid Chart Access

- Lucid for Education: <https://help.lucid.co/hc/en-us/articles/18624964476692-Welcome-to-Lucid-for-Education>
 - 3 editable LucidChart documents with a limit of 60 objects per document.
- Canvas now Includes Lucid Integration
 - Students can collaborate
 - <https://help.lucid.co/hc/en-us/articles/14559937492756-Integrate-Lucid-with-Canvas>

Share "VBA 3 Practice"

Individual access

Add people or Lucid groups

Edit and share ▾

Add: dzivomo22@gmail.com

Shareable link

On ☒

<https://lucid.app/lucidspark/42d839cb-c23e-4dd...>

Copy link

Anyone with the link ▾

Edit ▾

Anyone with the link

Any logged in user

Anyone at my organization

Manage

AR

BS

BB

CB

CA

CZ

KH

MR

PG

RC

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LucidChart/ LucidSpark

- Canvas - Move Lucid into your Navigation

Lucid

Welcome, Kellie

Through Canvas, teachers now receive a free license to the Lucid Education Suite!

The whiteboard has been at the center of the classroom for decades—for good reason. It's a convenient collaboration space where students and teachers can share ideas and organize thoughts.

Open Lucid ↗

Great Ideas

Offline access

Enhanced search

Personalized dashboards

Advanced filtering

Real-time analytics

Customizable notifications

Genification

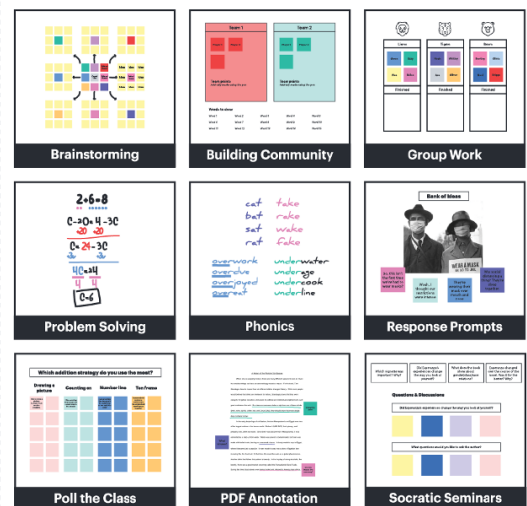
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* Lucidspark

Collaborative virtual whiteboard

- Brainstorming
- Building Community
- Group Work
- Problem Solving
- Response Prompts
- Poll the Class
- PDF Annotations

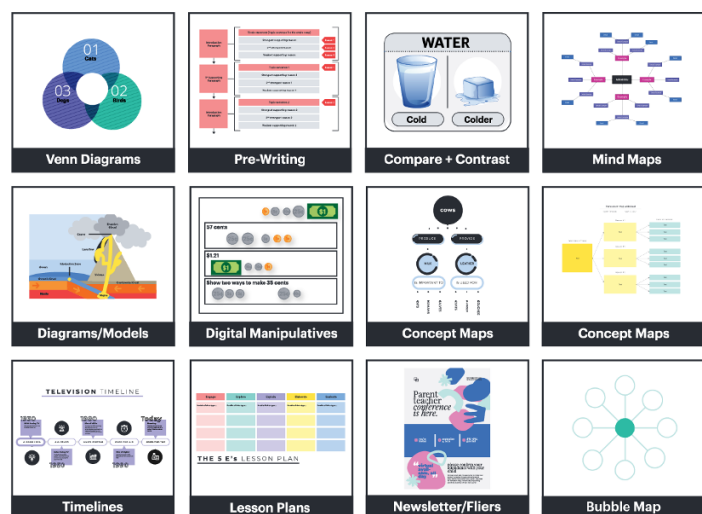


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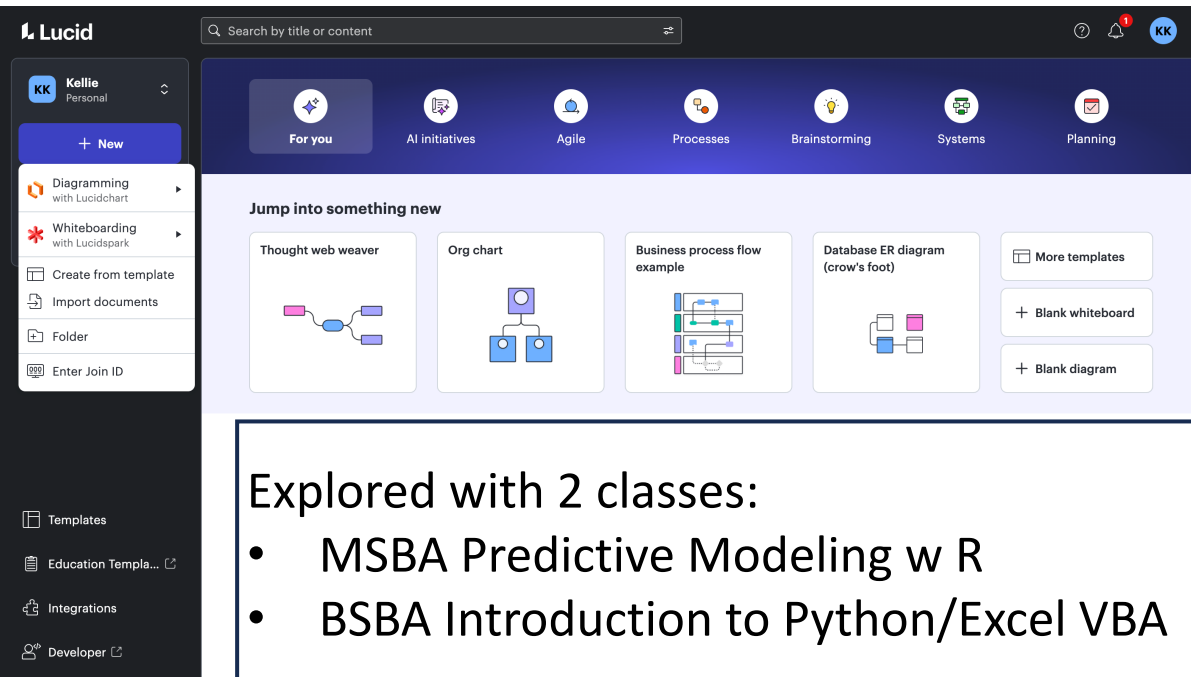
Lucidchart

Interactive diagramming and graphic organizers

- Venn Diagram
- Compare + Contrast
- Maps
 - Mind Map
 - Concept Map
 - Bubble Map
- Timelines



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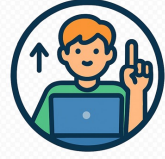


The screenshot shows the Lucid software interface. On the left is a sidebar with a user profile for 'Kellie Personal', a '+ New' button, and a list of templates including 'Diagramming with Lucidchart', 'Whiteboarding with Lucidspark', 'Create from template', 'Import documents', 'Folder', and 'Enter Join ID'. Below this are 'Templates', 'Education Templa...', 'Integrations', and 'Developer' links. The main area has a top navigation bar with icons for 'For you', 'AI initiatives', 'Agile', 'Processes', 'Brainstorming', 'Systems', and 'Planning'. Below this is a section titled 'Jump into something new' with four template cards: 'Thought web weaver', 'Org chart', 'Business process flow example', and 'Database ER diagram (crow's foot)'. To the right of these cards are buttons for 'More templates', '+ Blank whiteboard', and '+ Blank diagram'. A large text box at the bottom of the interface contains the following text:

Explored with 2 classes:

- MSBA Predictive Modeling w R
- BSBA Introduction to Python/Excel VBA

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 CONSTRUCTIVISM	 ACTIVE LEARNING	 METACOGNITION
• Learners build new knowledge by connecting ideas to their existing understanding. • Learning is most effective when students actively construct meaning through exploration, discussion, and reflection.	• Students engage directly with material through doing, discussing, and applying concepts. • Promotes deeper understanding by shifting students from passive listening to active participation.	• Students monitor and evaluate their own thinking, strategies, and understanding. • Strengthens learning by helping students plan, reflect on, and regulate their cognitive processes.


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Structured Thinking & Problem Solving

Procedural Programming Planning (Recipe)

- Provides a repeatable model for approaching assignments

Visualizing Programming Logic (Flowchart)

- Helps students break problems into steps
- Future!

Data Modeling for Information Systems (Entity–Relationship Diagrams - ERDs)

- Supports conceptual modeling before implementation
- Builds database literacy and abstraction skills



Visual scaffolds reduce cognitive load, making complex reasoning processes visible and learnable.



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Recipe

Inputs

- None

Outputs

- average sales summary for regions (singles or doubles)

Variables

- qtr12ws (first ws to hold first csv information)
- qtr34ws (second ws to hold second csv information)
- combined_ws (combined sales info)
- region_sales (total sales and count per region)
- summary (final output/worksheet)

Processing

1. Read SalesQtr12.csv into qtr12ws and SalesQtr34.csv into qtr34ws.
2. Append all rows from qtr12ws and qtr34ws into the new combined_ws list.
3. Create region_sales and set the total sales and count to zero for all four regions (East, West, South, North).
4. Iterate through every row in combined_ws and add the Sales24 value to the correct region's total and increment its count in region_sales.
5. For each region, divide the total sales by the count to determine the average sales.
6. Create summary rows using the region name and its newly calculated average sales.
7. Display the final summary worksheet.

• Make a macro-enabled workbook with a Menu sheet and two buttons: Import CSV and Value Counts.

• When Import CSV is clicked, ask for a CSV file, load it into a sheet called "Data," and confirm it's done.

• When Value Counts is clicked, ask for a range, count how many times each value appears, and show the results on a sheet called "ValueCounts."

• To combine two CSV files, ask for both files, copy their data into one sheet called "CombinedData," and confirm when complete.

• Save the workbook and test that all buttons work properly.

1. Save a macro enabled workbook and import files
2. Combine both files into a big table with all qtrs
3. group the data by region and for each region add all sales24 numbers and find average
4. create a ne table with two columns (region name, average sales value)
5. Put that table into excel with column A as Region, and column B as Average sales

Suppose you have the following 2 CSV files:

SalesQtr12.csv

Qtr	Region	Sales24
1	East	45
1	West	34
1	South	23
1	North	51
2	East	56
2	West	34
2	South	22
2	North	24

SalesQtr34.csv

Qtr	Region	Sales24
3	E	45
3	W	34
3	S	23
3	N	51
4	E	56
4	W	34
4	S	22
4	N	24

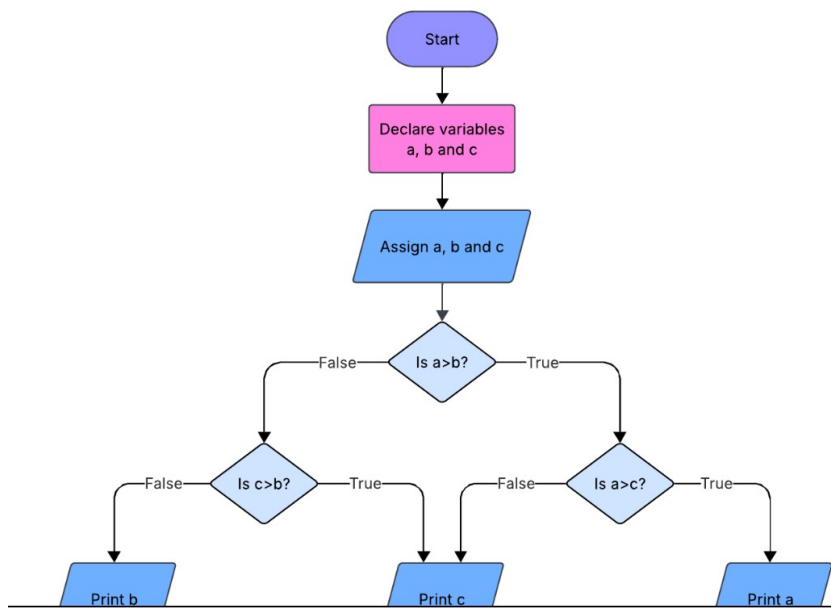
Create the following summary in Excel.

	A	B
1	Sales24	Average Sales
2	East	50.5
3	West	34.0
4	South	22.5
5	North	37.5

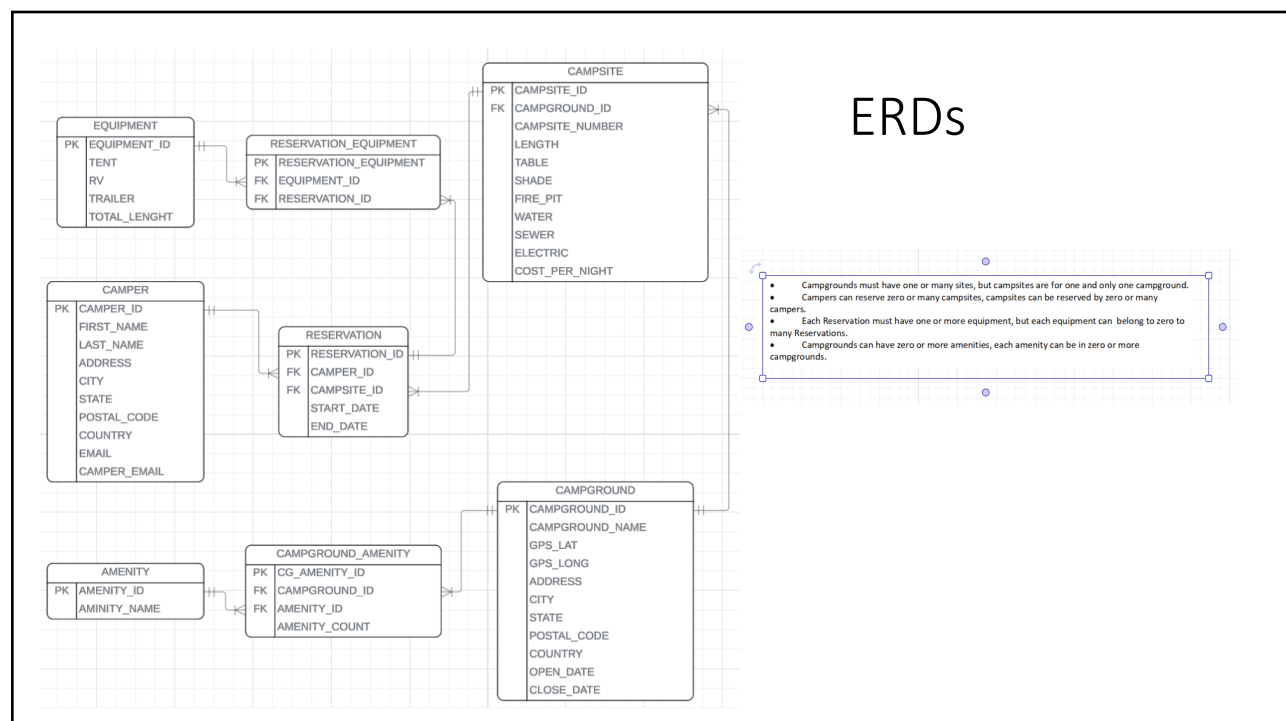
See all answers - compare

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FUTURE: FLOW CHART!



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Active Learning Pedagogy & Social Constructivism

Sticky Notes for Rapid Feedback

- Encourages participation from all students (safe, lower-barrier to share)
- Enables fast formative assessment

Collaborative Mind Maps

- Students co-construct meaning for course concepts
- Surfaces misconceptions and knowledge gaps

Peer Review & Decision-Making Tools (Voting)

- Builds analytic skills through comparison and critique



Interactive Classrooms where students learn by contributing and engaging with peers.

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Sticky Notes

Create answers to the questions based on your last name. Use a sticky note with the following at the top:
First Name with Last Initial
Favorite Food
Dream Vacation Location

presentations are on August 21, group tasks are self graded based on answers provided, the solutions will be posted on Canvas under Modules

- learn how to effectively deliver the results of the modeling exercises and projects in technical and nontechnical manners

• Market Processes
• Supply Chains
• There are three parts to the project: presentations are August 21st, group tasks are self graded, and the answers can be found under the modules after submission

A - D
How can you contact or make an appointment with me? When can you get help? Is there an attendance policy? What is Kurzweil 3000?

E - J
When can you submit an assignment due at 11:59 without a penalty? Can you submit work late? What is an oops ticket?

K - Q
Which is the most interesting outcome of this course to you?

R - Z
How many parts make up the project? What day are our project presentations? Are group tasks graded? If so, how? Where and when can you find solutions to the group tasks?

Enchiladas, Corny Islands, Should turn in an assignment by 8am next day for no late penalty. We have 1 oops ticket for project and 2 for group task. Late work is accepted with a penalty.

Stiracha, Maine! Submit by 8am the next day w no penalty. You can submit but with % penalties. You can have 1 oopsie that is a late submission w no penalty.

Butter Chicken
Somewhere in Europe maybe / won't count as late as long as it is in by 8a the following morning

Late work will be accepted with a penalty of 12 percentage points a day (.5% per hour) allow you turn in late work without an

Wings Japan
8am next day
Can submit work late
Oops ticket: can turn in assignment 1.5 weeks late

Streak dinner or pesto
Rio de Janeiro, Brazil
8am the following day = no penalty, 12%/day, you get one oops ticket to turn in a late assignment without penalty

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Sticky Notes



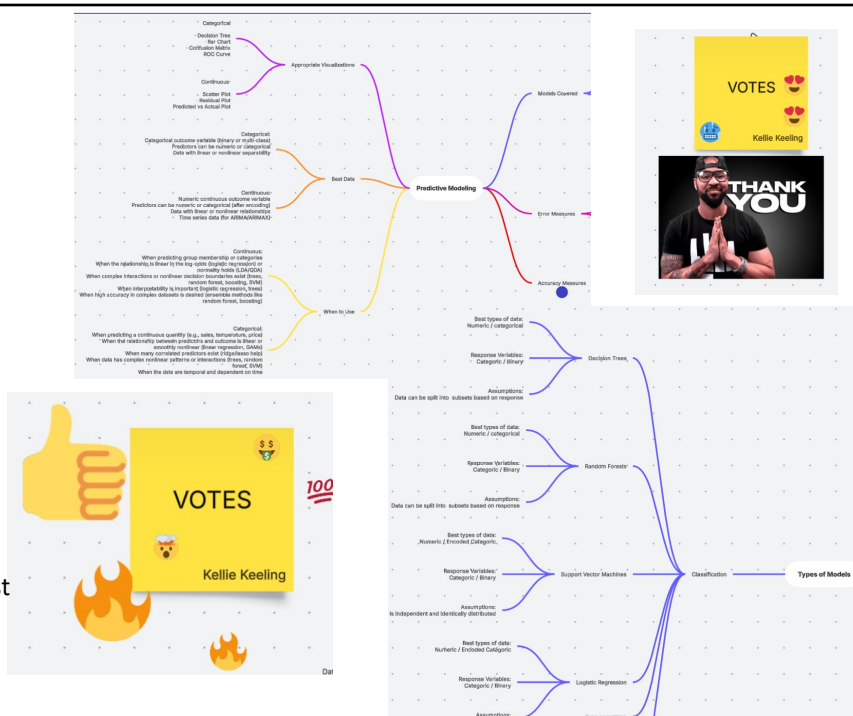
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Mind Maps w/ Voting

In class, share why

Last class you created a comparison of the models that we have covered in class. **Share your answers to the Comparison of Models** if you didn't work together in class. Create a mind map that shows how the models are similar and different in terms of response variables, assumptions, and best type data for and/or when to use. Use AI to help you make these comparisons.



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Enhance Analytical & Metacognitive Skills

Demonstrate Analytical Thinking (Present Results)

- Helps students articulate reasoning
- Encourages clear, organized communication of results

Comparative Concept Maps

- Builds conceptual understanding
- Encourages higher-order analysis (similarities/differences)

Reflective Learning Tools

- Using diagrams to explain or justify decisions “thinking about thinking”
- Makes learning processes more transparent



Students deepen understanding when they visualize how they think, compare, and assess ideas



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Present Analytic Results

Screenshots

Model Summary			
term	M7 Log(Dis)-Tax-Age-Nox	term	Model w/ log(dis), rm^2, & rm:lstat
(Intercept)	0.000	(Intercept)	0.000
crim	0.000	crim	0.000
zn	0.011	chasYes	0.001
indus	0.000	nox	0.000
chasYes	0.001	rm	0.005
rm	0.000	ptratio	0.000
ptratio	0.000	lstat	0.000
lstat	0.000	rad	0.000
rad	0.476	logdis	0.000
dis_log10	0.000	l(m^2)	0.000
ADJ R2	0.717	rm:lstat	0.000
		ADJ R2	0.812

- age - tax -
dis -indus -
zn

term	Pruned Model rm:lstat + zn:dis	term	log(medv) + interaction terms
(Intercept)	0.005	(Intercept)	0.000
crim	0.000	crim	0.000
zn	0.084	rm	0.000
chasYes	0.001	zn	0.339
nox	0.000	chas	0.006
rm	0.000	nox	0.000
dis	0.005	ptratio	0.000
ptratio	0.000	lstat	0.000
lstat	0.000	rad	0.000
rad	0.000	logdis	0.000
dis_log10	0.000	crim:rm	0.000
zn:dis	0.045	lstat:rad	0.000
rm:lstat	0.000	ADJ R2	0.800
ADJ R2	0.805		

term	Model w/ log(dis), No Age/Tax/Dis/Indus
(Intercept)	0.000
crim	0.000
zn	0.004
chasYes	0.000
nox	0.000
rm	0.000
ptratio	0.000
lstat	0.000
rad	0.007
log_dis	0.000
ADJ R2	0.740

term	reg_bos6v1_fit log10(medv)
(Intercept)	0.000
crim	0.000
zn	0.667
chasYes	0.000
nox	0.000
rm	0.000
ptratio	0.000
lstat	0.000
rad	0.008
log10(dis)	0.000
ADJ R2	0.778



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Present Analytic Results

Tables

A	The coefficients for Limit and Rating get larger and the R Squared decreases slightly. The AdjR ² is the same in both individual models. The coefficients change within the 3 models which impact their predictive powers (higher/lower coefficient estimates).
B	AR2 decreases slightly as we modeled a single dimension vs both. The dimensions coefficients increase when one is removed in both scenarios as compared against both Rating & Limit.
C	The R2 is the same for the models with removing just limit and removing just rating. The R2 slightly increases for the model with "limit and rating"
D	The Adjusted R Squared stays relatively the same including both predictors and then removing, but does slightly decrease by 0.002 when predictors are removed. Both coefficients increased when the other predictor was removed.
E	the coefficients increased by about double when one of the variable is omitted while the R2 decreased.
F	When one of the predictors is removed, the R-squared decreases just barely. Additionally, the coefficients for the remaining predictor increase when compared to their values in the model with them together.
When removing one of the predictors (rating, limit) the ADJ R squared only	

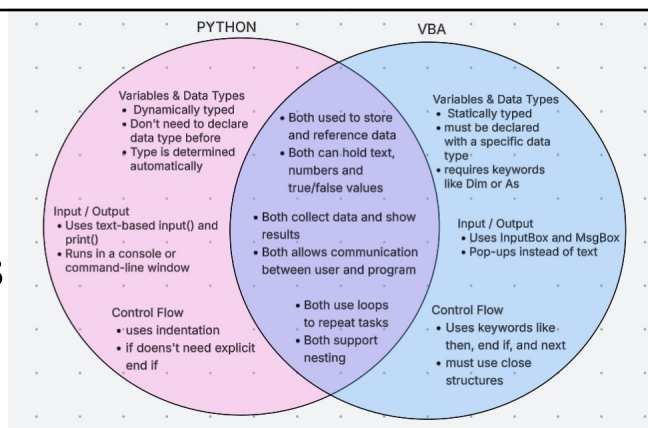
Group	Methods for Quant target (Regression)	Group	Metrics/Graphs for Quant target (Regression)
A	Linear Regression, Ridge/Lasso Regression, GAMs, GLMs	C	Residual plots + ↓
B	Random Forest, Decision Tree, k-nearest neighbor ???	D	MAE, RMSE, Adj. RSQ, predicted vs actuals plot

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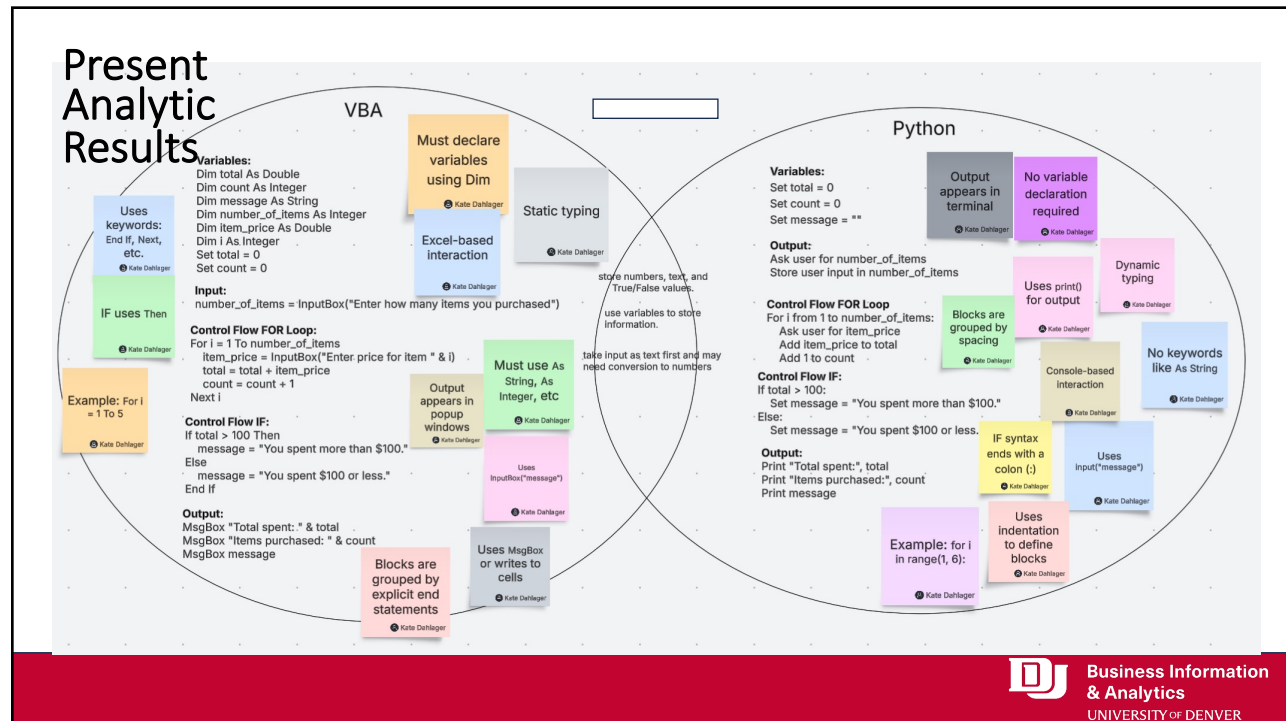
Venn Diagram Comparisons

Think about the following 3 programming concepts:

- **Variables & Data Types**
- **Input / Output**
- **Control Flow (IF and FOR Loops)**
- Find a place on the Lucid document to put your answer and put your names. For each category, using one of the diagramming tools, compare how the concept appears in **Python** and how it appears in **VBA**. What are the similarities or key differences between them?



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FUTURE

Reflective Learning Tools

- Using diagrams to explain or justify decisions "thinking about thinking"
- Makes learning processes more transparent

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Trick for Creating Force-Copy Template

Allows students to create new document based on your "template/question"

- Open your Lucid document.
- Find the shareable link for the document.
 - https://lucid.app/lucidspark/cf1eee35-2d21-4bc4-94c2-f36586617b55/editNew?page=0_0#
- In the URL, replace the word edit with editNew



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Lucid as a Multi-Dimensional Teaching Tool:

bit.ly/KKPresentationFiles

1. **Scaffolds Thinking** - helps students plan, model, and organize complex work
2. **Promotes Active Learning** – increases engagement and collaboration
3. **Builds Metacognition** – improves analytical reasoning and self-reflection



Data, Analytics, and Statistics Instruction (DASI)

DSI Nov 22 – 24, 2025 in Crystal G1

Bob Andrews, Virginia Commonwealth University

Kellie Keeling, University of Denver

Reza Kheirandish, Clayton State University

Email ideas for presentation at SEDSI to randrews@vcu.edu

SUNDAY

4:30-6 PM Panel Discussion of Current Issues and Challenges for Data-Driven Decision-Making Instruction

MONDAY

8:15-9:30 AM Panel Discussion of Current Issues and Challenges for Data-Driven Decision-Making Instruction

10-11:30 AM Preparing for AI-Augmented Decision Environ: Insights from Internships, Assessment, and Ethics

1-2:30 PM Building Responsible Fluency in Business Education

2:50-4:20 PM Business Meeting of participants in Data, Analytics and Statistics Instruction (DASI)

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